Week 5 Homework

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In order to receive full credit problems and solutions must be clearly written and presented neatly. In writing up your solution, you should state the problem as explaining the solution. Merely giving answers to exercises will is not enough to receive credit. Your write up must show work and convince me that you understand the material. If you need additional time to complete this assignment, please email me. Starred () proof problems are eligible for inclusion in the proof portfolio assignment.

Exercises

Exercise

What is oeis.org? What information can you find about sequences there? How might you use this resource for this course?

Exercise

An 8-person committee is to be formed from a group of 15 women and 12 men. In how many ways can the committee be formed if the committee must contain four men and four women? if there must be at least two men? if there must be more women than men?

Exercise

A **palindrome** is a string whose reverse is identical to the original string. Give examples of bit strings of length 5 and 6 that are palindromes. How many bit string palindromes of length n are there? How many ternary string palindromes of length n are there?

Exercise

How many strings of 6 letters of the english alphabet contain one vowel? exactly two vowels? at least one vowel? at least two vowels? What is number of strings of length n with exactly k vowels?

Exercise

A string that contains only 0s,1s, and 2s is called a **ternary string**. How many strings of 10 ternary digits (0, 1, or 2) are there that contain exactly two 0s, three 1s, and five 2s? Generalize this to the number of ternary strings of length k with a 0's, b 1's, and c 2's where $a + b + c = k$.

Exercise

How many integer solutions are there to the equation $x + y + z = 8$ for which:

1. x, y, z and are all positive?
2. x, y, z and are all non-negative?
3. x, y, z and are all greater than or equal to -3.

Exercise

How many ways are there to place 15 identical balls into 5 labeled boxes:

1. Each box has at least one ball?
2. Each box has at least two balls?
3. One box has exactly 10 balls?

Exercise

Suppose a professor has 10 identical calculators. How many ways are there to give these calculators to 3 students such that:

1. There are no other conditions?
2. Each student gets 1 calculator?
3. Each student gets 2 calculators?
4. Each student gets 3 calculators?

Exercise

Your favorite math themed fast food drive-in offers 20 flavors which can be added to your soda. You have enough money to buy a large soda with 4 added flavors. How many different soda concoctions can you order if:

You refuse to use any of the flavors more than once?
You refuse repeats but care about the order the flavors are added?
You allow yourself multiple shots of the same flavor?
You allow yourself multiple shots, and care about the order the flavors are added?

Proofs

Proof

Prove $\binom{n}{k}\binom{n-k}{l} = \binom{n}{l}\binom{n-l}{k}$ with $k + l \leq n$.

Proof

Let m and n be two integers. Prove that $mn + m$ is odd if and only if m is odd and n is even. Hint: You will need two subproofs. one for the if part and one for the only if part.

Proof

Prove that $n^2 + n + 1$ is odd for every integer n by direct proof.

Proof

Let n be a positive integer. Show that $\binom{2n}{n+1} + \binom{2n}{n} = \frac{1}{2}\binom{2n+2}{n+1}$.

Proof

Prove

$$\frac{1}{1 \cdot 4} + \frac{1}{4 \cdot 7} + \frac{1}{7 \cdot 10} + \cdots \frac{1}{(3n-2)(3n+1)} = \frac{n}{3n+1}$$

Proof

Prove the hexagon property:

$$\binom{n-1}{k-1}\binom{n}{k+1}\binom{n+1}{k} = \binom{n-1}{k}\binom{n+1}{k+1}\binom{n}{k-1}$$

Proof

Prove

$$\sum_{k\geq 0}\binom{r}{k}\binom{s}{n-k} = \binom{r+s}{n}$$

Hint: See Example 1.10 on page 9 of the course text.

Proof

Prove

$$\sum_{k\geq 0} k\binom{m}{k}\binom{n}{k} = n\binom{m+n-1}{n}$$

Hint: See Example 1.10 and 1.11 on page 9 of the course text.

Proof

Use induction to prove that $1+\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{3}}+...+\frac{1}{\sqrt{n}}\leq 2\sqrt{n}$ for all positive integers n.

Proof

Using induction, prove that $(\cos x + i\sin x)^n = \cos nx + i\sin nx$ for $n \geq 0$. Hint: Use the angle addition formulas.

Proof

Prove

$$\sum_{i=0}^{\lfloor n/2\rfloor}\binom{n}{2i} = 2^{n-1}$$

where $\lfloor x\rfloor$ is the floor function.